A Log-Linear IRT Accumulator Model of Early Word Learning

Draft specification (revised: 2PL extension)

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1 Research questions driving the modeling

The model is organized to answer three research questions. The baseline is the minimal specification that lets each be posed and answered; each variant in the nested family (§11) sharpens one of them. A fourth question — when does accumulation start? — is posed in the appendix because the start-time parameter s is near-non-identifiable in practice (see §A).

RQ1. Is there acceleration in the accumulation process? McMurray (2007) and Mitchell & McMurray (2009) argued from theory that the empirical "vocabulary explosion" requires acceleration in the threshold distribution $g(\tau)$: pure constant-rate accumulation produces growth that's too smooth, and leveraged learning shifts the curve in time but does not create acceleration that isn't already present. Bergelson (2020) made the empirical case: older infants are "better learners." We formalize this question via δ: in our linear predictor, time enters as $(1+\delta)\log((t-s)/a_0)$, where $\delta = 0$ corresponds to McMurray's pure linear accumulator (one log-token of evidence per log-age unit, no acceleration). Posterior $\delta > 0$ with CrI excluding zero is a quantitative confirmation of the McMurray–Mitchell acceleration claim.

RQ2. How well does frequency reconstruct word difficulty? Braginsky et al. (2019) showed frequency is only modestly related to AoA within lexical classes. We test this within our model by asking how much of word-level residual difficulty ψ_j is explained by $\log p_j$, and whether the per-class slope β_c on $\log p_j$ deviates from the unit-accumulator baseline $\beta_c = 1$.

RQ3. How much of child-level outcome variance is input-driven? This is the σ_α question: once we pin the variance of CDS input rate r_i from external data, how much residual child variance must be absorbed by a learning-efficiency term α_i? Posterior on $\pi_\alpha = \sigma_\alpha^2/(\sigma_r^2 + \sigma_\alpha^2)$ reports the efficiency share of intercept variance.

Each RQ maps onto a specific posterior quantity, recovered as the nested family is built up:

2 Motivating the model: how much variation is there to explain?

Before fitting any accumulator model, it is worth asking how much between-child variation in language ability actually exists in the data, and how that variation relates to age-driven

RQ	Quantity	How answered
1	δ	Posterior in M2 vs. M1; baseline (M1) pins $\delta = 0$.
2	β_c, $r(\hat{\psi}_j, \log p_j)$	Posterior in M4 (per-class β_c); residual correlation in M3.
3	$\pi_\alpha = \sigma_\alpha^2/(\sigma_r^2 + \sigma_\alpha^2)$	Derived posterior in M3+; σ_r^2 pinned externally.

Table 1: Mapping from research questions to model quantities and nested-family stages (M1–M5).

development. We ran the simplest possible 1-PL IRT — $P(y_{aj} = 1) = \sigma(\theta_a - \beta_j)$, with one ability θ_a per admin (*not* per child) and one difficulty β_j per word — fit via `lme4::glmer` on the Wordbank English longitudinal sample. No time term, no frequency, no per-child structure. Just: how variable is ability across admins, and how does the population mean change with age?

Two reference quantities. The population-mean θ rises by $\hat{\beta}_{\text{age}} = 0.47$ logits/month on this sample (a regression of admin-level θ_a on age). Pooled across all admins, $\text{SD}(\theta_a) = 3.25$ logits; the residual within-age SD (after subtracting the linear age trend) is 1.95 logits.

Translating SD into months of development. A natural unit for individual differences is "how many months of typical development does ±1 SD across kids correspond to?" On this sample,

$$\frac{\text{SD}_{\text{within-age}}(\theta)}{\hat{\beta}_{\text{age}}} = \frac{1.95}{0.47} \approx 4.2 \text{ months of development per SD.}$$

A child one SD above the within-age mean looks, on the IRT scale, like a child of average ability 4 months older.

Variation grows with development. The within-age SD is not constant: it rises from ~ 1.0 logit at 12–14 mo ($\sim$ 2 months equivalent) to ~ 2.7 at 26–30 mo ($\sim$ 5.8 months equivalent). Children *fan out* rather than converge as they develop — a finding that any model of individual differences must accommodate.

What this implies for the model. The variation we need to explain is large and increasing with age. A pure accumulator with constant per-child input rate cannot produce growing spread; we need either age-varying efficiency (per-child slope deviation ζ_i) or some other mechanism. The remainder of the explainer builds the nested model family that makes structural choices about *which* mechanism. Figures `precursor_trajectories.png` and `precursor_variation.png` (in `model/figs/longitudinal/`) and the script `model/scripts/precursor_irt.R` reproduce these numbers.

3 One-paragraph summary

A child i of age a_i has, on average, heard word j approximately $T_{ij} = r_i\, p_j\, H\, (a_i - s)$ times, where r_i is the child's tokens-per-hour of CDS, p_j is the word's corpus probability, H is waking hours per month, and s is the global start time for accumulation. The child produces word j on the CDI with probability $\sigma(\log T_{ij} + \log \alpha_i - \psi_j)$ under a canonical Rasch/IRT logit link ($\sigma(x) = 1/(1 + e^{-x})$), where α_i is the child's learning-efficiency multiplier and ψ_j is the word's log-threshold.

Every quantity lives on a log-token scale with $O(1)$ priors, so Stan is well-conditioned. The crucial move is that r_i's population distribution is pinned by external data (Sperry/HR/WF), so σ_α is identifiable and Question 4 has an answer. Questions 1–3 are posterior summaries of the baseline; planned variants sharpen them.

4 Quantities

Symbol	Meaning	Units
r_i	Child i's CDS input rate	tokens / hour
α_i	Child i's learning-efficiency multiplier	dimensionless
p_j	Word j's corpus probability (CHILDES)	dimensionless, $\sum_j p_j = 1$
H	Waking hours per month (fixed)	hours / month
a_i	Age at CDI assessment	months
s	Global start time (accumulation onset)	months
ψ_j	Log expected tokens of word j to acquisition	log-tokens
δ	Age rate-change exponent	dimensionless
λ_j	Word j's 2PL discrimination	dimensionless (> 0)
a_0	Reference age (fixed)	months
y_{ij}	Child i produces word j on CDI	$\{0, 1\}$

Fixed constants. $H = 12 \times 30.44 \approx 365$ hours/month (12 waking hours/day). $a_0 = 20$ months.

5 Generative model

5.1 Linear predictor

For child i, word j, age $a_i > s$:

$$\eta_{ij} = \lambda_j \cdot \Big[\log r_i + \log \alpha_i + \log p_j + \log H + (1+\delta)\, \log\Big(\tfrac{a_i - s}{a_0}\Big) - \psi_j\Big]. \tag{1}$$

Here $\lambda_j \geq 0$ is a word-specific *discrimination* parameter (2PL). With $\lambda_j \equiv 1$ the model collapses to the Rasch form $\eta_{ij} = \log(r_i \alpha_i p_j H(a_i - s)) - \psi_j$ — the log ratio of tokens heard to tokens needed, with unit logit-per-log-token gain. With λ_j free, each word has its own gain; high λ_j produces a sharp transition near ψ_j, low λ_j a gradual one. Section 7 motivates the extension in terms of the accumulator and in terms of model fit.

5.2 Likelihood

Canonical Rasch/IRT logit link:

$$y_{ij} \mid \eta_{ij} \sim \text{Bernoulli}\big(\sigma(\eta_{ij})\big), \qquad \sigma(x) = \frac{1}{1+e^{-x}}. \tag{2}$$

The logit link is the standard "log-linear IRT" choice and is numerically well-behaved in Stan. ψ_j is interpretable as the log-threshold in log-tokens on the logit scale (the scale factor is absorbed implicitly; $\exp(\psi_j)$ is proportional to, not exactly, the expected number of tokens to 50%-acquisition).

Edge case: when $a_i \leq s$, set $\mathbb{P}(y_{ij} = 1) = \varepsilon$ and exclude these cells from the log-age term (in the code this is implemented by flooring $a_i - s$ at 0.01 inside the log).

5.3 Priors

Child-level latent (collapsed). The likelihood depends on $\log r_i$ and $\log \alpha_i$ only through their sum, so we sample a single child-level latent

$$\xi_i := \log r_i + \log \alpha_i \sim \mathcal{N}\big(\mu_r,\ \sigma_r^2 + \sigma_\alpha^2\big).$$

Here $\mu_r = \text{mean}(\log r)_{\text{Sperry pool}} \approx 7.34$ (geometric mean ≈ 1537 tokens/hour) and $\sigma_r^2 = \text{Var}(\log r)_{\text{Sperry pool}} \approx 0.534^2$ are *pinned* from external data (this is the identifiability lever for RQ4). The pool is the $n = 42$ dyads with non-NA `adult_child_tokens_hr` in `data/raw_data/sperry/hourly_tok` in practice these are all from Sperry et al. (2019), since Hart & Risley (1995) and Weisleder & Fernald (2013) reported only mother-only and all-adult counts, respectively. See `input_estimation/` for the full validation set and a sensitivity-range justification. σ_α is free:

$$\sigma_\alpha \sim \text{HalfNormal}(0, 1). \tag{3}$$

The per-child $\log \alpha_i$ is a *derived* quantity with closed-form posterior conditional on ξ_i:

$$\log \alpha_i \mid \xi_i \ \sim\ \mathcal{N}\left(\frac{\sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2}\,(\xi_i - \mu_r),\ \frac{\sigma_r^2 \sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2}\right),$$

reported in generated quantities. (Rationale for this collapsed form is in §6: sampling both latents individually over-parameterizes the model and creates bad posterior geometry in HMC.)

Word thresholds by lexical class.

$$\psi_j \mid \text{class}(j) = c \sim \mathcal{N}(\mu_c, \tau_c^2), \quad \mu_c \sim \mathcal{N}(8, 3^2), \quad \tau_c \sim \text{HalfNormal}(0, 1). \tag{4}$$

Note on μ_c. The earlier value $\log 1000$ was a back-of-envelope centering. A more principled choice: tokens heard of a typical word by age 20 mo spans roughly e^5 (rare words) to e^{12} (function words), so $\mu_c \sim \mathcal{N}(8, 3^2)$ is weakly informative across the relevant range and dominated by the data.

Global start time (RQ2).

$$s \sim \mathcal{N}(4.5, 2^2) \text{ truncated to } [0, 15]. \tag{5}$$

One s for the whole model, per your decision. The prior is centered at 4.5 mo — M&P's comprehension estimate was ∼2 mo; production is likely later. s close to 0 would indicate accumulation begins near birth.

Age rate-change (RQ3).

$$\delta \sim \mathcal{N}(0, 0.5^2). \tag{6}$$

Word discrimination (2PL).

$$\log \lambda_j \sim \mathcal{N}(0, \sigma_\lambda^2), \quad \sigma_\lambda \sim \text{HalfNormal}(0, 1). \tag{7}$$

Setting $\sigma_\lambda \to 0$ (via a near-zero prior SD passed as data) pins every $\lambda_j = 1$ and recovers the Rasch model. This is the toggle used in code: one Stan file serves both Rasch and 2PL variants. Scale identifiability (the classic $\lambda_j \leftrightarrow 1/\lambda_j$, $\psi_j, \xi_i \leftrightarrow \lambda_j\psi_j, \lambda_j\xi_i$ ambiguity of 2PL) is resolved by our pinning of σ_r: the ξ scale is fixed externally, so ψ_j and λ_j are separately identified.

$\delta = 0$ is the pure cumulative null. $\delta > 0$ indicates older children accumulate faster — a formal test of the Bergelson comprehension-boost claim.

6 How each RQ is answered

RQ1 — is there acceleration?

Single parameter δ, entering the time term as $(1 + \delta) \log((t - s)/a_0)$. $\delta = 0$ is McMurray's pure linear accumulator: each log-age unit contributes one log-token of evidence. $\delta > 0$ means evidence accumulates super-linearly in log-age (or equivalently: tokens grow as $t^{1+\delta}$ on the linear scale, faster than the linear-accumulation baseline). Headline posterior: $\hat{\delta}$ with CrI; CrI excluding zero is a quantitative confirmation of the McMurray–Mitchell argument that acceleration is necessary to produce empirical vocabulary growth.

The natural ablation is `pin_delta=0` (M2 vs. M1 in the nested family, §11), which forces the model back to the pure accumulator. If δ is doing real work, this ablation should produce a substantially worse fit and force the other parameters into pathological regions.

RQ2 — does frequency reconstruct word difficulty?

The Rasch baseline (M1 in §11) sets the coefficient on $\log p_j$ to 1 (the pure-accumulator assumption: each token of word j contributes one log-token of evidence). Posterior $\hat{\psi}_j^{(s)}$ is then residual word difficulty after unit-coefficient frequency. We report:

- Posterior correlation $r(\hat{\psi}_j, \log p_j)$ and R^2 — what fraction of word-level difficulty is explainable by frequency.
- Posterior class means $\hat{\mu}_c$ — does class structure persist after frequency is accounted for?
- Posterior β_c in M4 (per-class slope on $\log p_j$) — do classes differ in how strongly frequency contributes?

The class-specific slope variant M4 is the principal RQ2 test. $\beta_c = 1$ corresponds to the unit-accumulator baseline; $\beta_c < 1$ means frequency contributes less per log-unit; $\beta_c \approx 0$ means frequency is uninformative once per-word ψ_j is free.

RQ3 — input- vs. efficiency-driven variance

The linear predictor contains $\log r_i + \log \alpha_i$ as a sum, unidentifiable from Wordbank alone. We break the degeneracy by pinning σ_r^2 at its externally-measured value (from Sperry, Hart & Risley, and Weisleder & Fernald pooled). The child-level latent $\xi_i = \log r_i + \log \alpha_i$ is then identified

from the data, with implied distribution $\xi_i \sim \mathcal{N}(\mu_r,\, \sigma_r^2 + \sigma_\alpha^2)$. Subtracting the pinned σ_r^2 from the inferred $\mathrm{Var}(\xi_i)$ identifies σ_α^2, giving

$$\pi_\alpha \;\coloneqq\; \frac{\sigma_\alpha^2}{\sigma_r^2 + \sigma_\alpha^2} \tag{8}$$

as a well-defined derived posterior quantity: the proportion of child-level outcome variance attributable to learning efficiency rather than input rate. Its complement, $1 - \pi_\alpha$, is the input-driven fraction.

What π_α does and does not decompose. The full between-child variance in θ_{it} at age t is

$$\mathrm{Var}_i(\theta_{it}) = \sigma_\xi^2 + \log_{age}^2\, \sigma_\zeta^2 + 2\,\rho_{\xi\zeta}\,\sigma_\xi\,\sigma_\zeta\,\log_{age}\,.$$

At $t = a_0$ (the pivot, where $\log_{age} = 0$), this collapses to σ_ξ^2 and π_α is exactly the efficiency share. Off-pivot, the per-child slope variance σ_ζ^2 contributes increasingly. With our fitted $\sigma_\zeta \approx 3.5$ on English, at $\log_{age} = -1$ (e.g., 7 mo when $a_0 = 19$) the slope component contributes $\sim 5\times$ the intercept variance.

We do not decompose σ_ζ into input-rate-growth and efficiency-growth components, because the model treats $\log r_i$ as age-invariant (see "Age-invariance of $\log r_i$" in § Assumptions). ζ_i therefore acts uniformly on $(\log r_i + \log \alpha_i)$ — without longitudinal observations of $r_i(t)$ within child, the data cannot separate it. We report π_α as the intercept-share at the pivot age, and treat σ_ζ as a third axis of between-child variation that exists alongside the input/efficiency intercept decomposition.

The `drop_slopes` ablation ($\sigma_\zeta = 0$) tests whether the intercept-level π_α is robust: forced into a single-locus model, the data still gives $\pi_\alpha \approx 0.92$ on English (vs. 0.91 with slopes free), so the headline number is robust to slope-vs-no-slope choices.

Why the collapsed parameterization matters practically. Sampling $\log r_i$ and $\log \alpha_i$ individually — each with its own non-centered $\mathcal{N}(0,1)$ prior and a scale parameter — leaves the posterior with a flat ridge: any rotation of the pair that preserves ξ_i has the same likelihood. Only the priors break the degeneracy, at the cost of 2I parameters where I are identified. In HMC this manifests as chains getting stuck in bad regions (we observed one chain running 28+ minutes on a sim the others cleared in 3). Sampling ξ_i directly, with σ_α entering only through $\sigma_{\xi_i}^2 = \sigma_r^2 + \sigma_\alpha^2$, eliminates this geometry entirely. Parameter recovery on a 250-child $\times$ 150-word synthetic dataset: π_α posterior 0.62 [0.50, 0.72] against true 0.61.

Sensitivity analysis. Refit with $\sigma_r^2 \in \{0.5, 1, 2\} \cdot \sigma_r^{2,\mathrm{obs}}$. Stability of π_α across these indicates how much the claim depends on trusting the external estimate.

7 Why 2PL: one-token-one-drop is too strong

The Rasch version of (1) (i.e. with $\lambda_j \equiv 1$) implies every word has the same logit-per-log-token gain. For a fixed word, the predicted slope of $\mathrm{logit}\, p_{ij}$ with respect to age is $(1+\delta)/(a_i - s)$ — on the order of 0.08 logits per month at typical observed ages. Real CDI words transition from $\sim$5% to $\sim$95% acquired over roughly six months (about 6 logits), which is $\sim 12\times$ faster than the Rasch slope can produce.

When we fit Rasch to Wordbank (500 children $\times$ 200 items), the posterior finds this mismatch in a predictable way: it cranks every available steepening lever — δ rises to ~ 2.3, s pushes to ~ 12 months, and crucially σ_α inflates to ~ 2 — implying children differ by $\sim 50\times$ in learning efficiency, and $\pi_\alpha \approx 0.93$. This pattern is stable across the s/δ identifiability, suggesting misspecification rather than genuine individual-difference variance.

The 2PL fix. Adding word-specific discrimination λ_j gives each word its own slope around threshold: the per-word logit derivative becomes $\lambda_j(1+\delta)/(a_i - s)$. Words whose referents are highly diagnostic of their meaning ("ball" when you see a ball; concrete nouns generally) can have large λ_j and transition sharply; words whose meaning is more distributional or context-dependent (function words, abstracts) have small λ_j and gradual transitions. This is the standard two-parameter logistic IRT extension and is strongly preferred over 1PL for CDI data in prior work (Kachergis, Marchman, Dale, Mankewitz, & Frank, 2022, *JSLHR*).

Three accumulator interpretations of λ_j (all observationally equivalent from CDI alone).

1. **Information per token.** λ_j is the per-token shift in the child's certainty about word j. High λ_j: each exposure is diagnostic.
2. **Sharpness of the cognitive threshold.** λ_j is the inverse width of the band over which evidence must accumulate for acquisition to occur.
3. **ELI-to-token ratio (bridge to Mollica & Piantadosi).** If only a fraction of heard tokens are effective learning instances and the rest are background noise, λ_j is proportional to that fraction for word j.

Falsification checks. The hierarchical prior on $\log \lambda_j$ (pooled via σ_λ) regularizes: one unusual word can't take over. Two further checks protect against freely flexible fit-to-noise:

1. σ_α should *decrease substantially* under 2PL. If the Rasch inflation was misspecification dressed as person variance, 2PL should absorb it into λ_j and drop σ_α toward a more reasonable value (e.g. $\sim$0.7), sharpening our answer to RQ4.
2. λ_j should correlate with measurable word properties: higher for concrete nouns, lower for function words and abstract adjectives. If λ_j is uncorrelated with any external measurement, we have over-fit rather than learned.

What this changes for the RQs. RQ1 (frequency-vs-difficulty) unchanged: we still report the posterior correlation of ψ_j with $\log p_j$. RQ2 (start time), RQ3 (δ) unchanged at the structural level, but the extreme posteriors observed under Rasch should moderate. RQ4 (variance decomposition) sharpens substantially: π_α becomes the proportion of child variance after item-level discrimination has been accounted for, which is the quantity we really want.

8 Assumptions, listed

1. **Pure accumulation.** Exposure accumulates linearly at a per-token rate independent of prior knowledge. No leveraged learning (Mitchell & McMurray 2009 showed leverage cannot

create acceleration that is not already present).

2. **Per-word evidence gain λ_j.** In 2PL mode each token contributes λ_j units of logit-scale evidence; $\lambda_j = 1$ reduces to the Rasch "one token, one drop" rule. Per-child gain is *not* modeled: all children receive the same λ_j for a given word. This is relaxable by adding a random slope, which we defer.

3. **Logit link with fixed scale.** Canonical Rasch/2PL IRT. Equivalent (up to a ~ 1.7 scale factor) to a probit model; approximates the Gamma cumulative of Mollica & Piantadosi for typical parameter values.

4. **Cross-sectional data.** Each child observed once.

5. **CDS only.** Overheard speech ignored in v1.

6. **Same input distribution as reference studies.** Our CDI sample has the Sperry/HR/WF distribution of r_i; SES moderators not modeled in v1.

7. **Age-invariance of $\log r_i$.** The model treats each child's input rate as a single per-kid constant, not a function of age. The per-child age-acceleration $(1 + \delta + \zeta_i)$ acts on the combined product $r_i \cdot \alpha_i$, so any age-varying child-side dynamics (efficiency growth, parents talking more to older kids) are absorbed into $\delta+\zeta_i$ without attribution. Empirically supported by within-kid regressions of $\log r_{iv}^{\text{obs}}$ on age in BabyView ($N = 20$ kids, 5688 videos: mean within-kid slope -0.006 logits/mo, median -0.002) and SEEDLingS ($N = 44$ kids, 525 LENA recordings: mean -0.011 logits/mo). Both are at least an order of magnitude smaller than the $(1 + \delta)\ \log_{age}$ effect over the same age range, so σ_ζ is unlikely to be primarily absorbing input-rate growth. See `model/scripts/input_rate_vs_age.R`.

8. **Global s.** One start time for all words. Per-class or per-word s is a planned extension.

9. **CHILDES frequencies as truth.** Sampling noise in p_j ignored; defensible for common words.

10. **Lexical class only via thresholds.** Class enters through the hierarchical prior on ψ_j; class-specific frequency slopes or start times are variants, not baseline.

9 Sampling strategy: children and items

The data we fit is large enough that we always work on a subsample. The strategy is the same across pipelines (Wordbank longitudinal, BabyView, future Seedlings) so results are directly comparable.

Children.

1. Filter to children with ≥ 2 admins (longitudinal coverage).
2. Compute each child's *median admin age* as their representative age.
3. Bin children into $N_{\text{age}} = 4$ age quantile bins.
4. Within each bin, take the top kids by number of admins, then by age span. (Tie-breaking on richer longitudinal data.)

5. Top up uniformly at random if any bin is short (rare).

This replaces an earlier, biased "top-by-priority" rule that just maximized $n_{\text{admin}} + \text{span}/5$; that rule overweighted data-rich kids in narrow age windows. The age-stratified rule guarantees coverage across the whole 16–30 mo developmental window.

Items.

1. Filter to items with non-zero CHILDES probability p_j.
2. For each lexical class, bin items into $N_{\text{diff}} = 4$ difficulty quartiles based on $\log p_j$.
3. Sample uniformly within each (class, difficulty-quartile) cell — 16 cells for 4 classes $\times$ 4 quartiles.
4. Top up at random if any cell is short.

The earlier item-sampling rule stratified only on lexical class, which oversampled high-frequency words (those are dense in the CHILDES distribution). The class-by-difficulty grid ensures we have hard, medium, and easy words within *each* class, which sharpens the identification of ψ_j and β_c when they enter as parameters.

Pipeline-shared item pool. All Wordbank-based pipelines (cross-sectional and longitudinal, English and Norwegian) draw from the *union of WG and WS forms*, mirroring the BabyView treatment. Each (child, age, form) is its own admin; a child who took both WG and WS at different ages contributes both. This extends English age coverage from the WS-only 16–30 mo to 8–30 mo and matches the cross-pipeline data definition.

Per-pipeline application.

- **Wordbank longitudinal (English / Norwegian).** 200 children $\times$ 200 items per dataset, drawn via the stratified sampling above. English: median 3 admins, 14 mo span; Norwegian: median 8 admins, 14 mo span (Norwegian has the much denser Kristoffersen contributor).
- **Wordbank cross-sectional (English).** 500 children, treated as the most recent (child, form) admin from `long_items.rds`. Stratified the same way as the longitudinal pipeline.
- **BabyView.** 20 longitudinal English subjects (the entire available pool) $\times$ 200 items, with input rate observed via 5 688 videos. No child-side subsampling: we use everyone.

10 What we deliver

1. **RQ-addressed posteriors**: δ (RQ1), β_c and $r(\hat{\psi}_j, \log p_j)$ (RQ2), and π_α (RQ3) — all with credible intervals.
2. **Per-word** ψ_j: absolute log-token thresholds for all 680 CDI items.
3. **Per-child** $\log \alpha_i$: child-level efficiency estimates.
4. **Counterfactual simulator**: given hypothetical (r, α, a), predicted vocabulary size with CrI.

5. **Planned model comparisons**: LOO-CV across the baseline plus variants for each RQ (see §8).

11 Nested model family

We build the model as a strictly nested progression from the simplest possible per-word IRT (M0) up to the full-feature variant. Each step adds one named component and answers one specific question; comparing adjacent models (LOO-CV / WAIC, posterior shifts on the structural parameters) attributes the marginal value of that component.

The model classes $c \in \{1, \ldots, C\}$ of word j (lexical category: nouns / predicates / function words / other) enter only through hierarchical priors on ψ_j (class-mean μ_c, class-SD τ_c) and, in M4, through class-specific frequency slopes β_c. Throughout, $\xi_i = \log r_i + \log \alpha_i$ is the per-child intercept with prior $\xi_i \sim \mathcal{N}(\mu_r, \sigma_r^2 + \sigma_\alpha^2)$ and σ_r externally pinned.

M0 — minimal IRT (no time, no frequency).

$$\eta_{ijt} = \xi_i + \log H - \psi_j.$$

Per-word ψ_j absorbs all between-item variance; per-child ξ_i absorbs all between-child intercept variance. No age effect, no frequency. This is the absolute null: vocabulary trajectories are flat in age, and word difficulty is a free per-word parameter only.

M1 — McMurray null (add unit-coefficient frequency and unit-rate accumulation).

$$\eta_{ijt} = \xi_i + \log \tfrac{t-s}{a_0} + \log H + \log p_j - \psi_j.$$

Adds the structural accumulator content: frequency enters with unit coefficient (one log-token of evidence per token of word j heard), and the time term grows at unit rate per log-age. This is McMurray (2007)'s pure linear accumulator. The slope on $\log_{age}$ is fixed at 1, with no per-child deviation. **M1 vs. M0 quantifies whether frequency and time contribute beyond per-word ψ_j and per-child ξ_i** (RQ2 partial answer).

M2 — add population acceleration δ (RQ1).

$$\eta_{ijt} = \xi_i + (1 + \delta)\, \log \tfrac{t-s}{a_0} + \log H + \log p_j - \psi_j, \quad \delta \sim \mathcal{N}(0,\, 0.5^2).$$

Frees the $\log_{age}$ slope as a deviation δ above the pure-accumulator baseline. $\delta = 0$ recovers M1. **M2 vs. M1 is the RQ1 test**: does the data require super-linear-in-age token accumulation?

M3 — add per-child slope deviations ζ_i.

$$\eta_{ijt} = \xi_i + (1 + \delta + \zeta_i)\, \log \tfrac{t-s}{a_0} + \log H + \log p_j - \psi_j.$$

Each child has their own deviation ζ_i from the population slope. Joint per-child prior: $(\xi_i, \zeta_i) \sim \mathrm{MVN}(0, \Sigma)$ with LKJ on correlation; sum-to-zero centering separates δ from mean(ζ_i). Requires longitudinal data for σ_ζ to be identified. **M3 vs. M2 measures between-child variation in growth rate.**

M4 — add class-specific frequency slope β_c (RQ2).

$$\eta_{ijt} = \xi_i + (1 + \delta + \zeta_i)\, \log \tfrac{t-s}{a_0} + \log H + \beta_{c(j)}\, \log p_j - \psi_j, \quad \beta_c \sim \mathcal{N}(1,\, 0.5^2).$$

Frequency now enters with class-specific weight. $\beta_c = 1$ is M3 (unit accumulator); $\beta_c = 0$ means frequency is uninformative for class c once per-word ψ_j is free. **M4 vs. M3 is the RQ2 test**: do classes differ in how strongly frequency contributes?

M5 — add per-word discrimination λ_j (2PL).

$$\eta_{ijt} = \lambda_j \Big[\xi_i + (1 + \delta + \zeta_i)\, \log \tfrac{t-s}{a_0} + \log H + \beta_{c(j)}\, \log p_j - \psi_j\Big], \quad \log \lambda_j \sim \mathcal{N}(0,\, \sigma_\lambda^2).$$

Per-word multiplicative scaling on the $(\theta - \beta)$ gap. $\sigma_\lambda \to 0$ recovers M4 (Rasch). **M5 vs. M4 measures item-level sharpness variance.**

Robustness variants (off the main spine).

Variant	Compared to	What it tests
`lmm`	M3	Linear-in-age ($(\beta_{age} + \zeta_i)(t - a_0)$) instead of log-linear; tests functional-form sensitivity.
`no_class`	M3	Drops class hierarchy on ψ_j; tests whether class structure adds anything.
`free_s`	M3	Frees s; tests s/δ trade-off (see Appendix A).
`no_freq`	M2	Drops $\log p_j$; tests whether frequency adds beyond per-word ψ_j.

The variant grammar implements the family. `variant_hyperpriors()` in `model/R/helpers.R` maps a variant name onto the prior overrides that select a specific model in the family. Tight priors near zero pin a parameter off (e.g., $\sigma_\zeta \sim \mathcal{N}(0, 0.001)$ pins $\zeta_i \equiv 0$); loose priors free it. The running log of fits is in `notes/experiments.md`.

12 Datasets and observation channels

The variant grammar (§11) toggles *model components*. Independent of that, each dataset comes with its own *observation channels* that determine which auxiliary parameters are appropriate. We separate the two levers in code so they can't be mixed by accident.

Lever 1 — variant grammar (per-fit toggles). Tight-prior switches on σ_λ, σ_ζ, s, δ. Implemented in `variant_hyperpriors()` (`model/R/helpers.R`); identical across pipelines (the `long_`, `io_` prefixes are stripped inside).

Lever 2 — dataset-level design choices (per-bundle constants). Properties inherent to the observation channel of a given dataset. Baked into `stan_data` by the `prepare_*.R` script; not a choice a researcher makes per fit. Two examples currently:

- *Reactivity inflation β_{react}.* Per-child observed input rate $\log r_{iv}^{\text{obs}}$ for video v of child i is $\log r_i + \beta_{\text{react}} + \varepsilon_{iv}$. The bias β_{react} absorbs systematic differences between observed input and "true" input.

- **BabyView** (head-mounted video, on-camera observer effect): $\beta_{\text{react}} \sim \mathcal{N}(0.4, 0.4)$ — active, with the prior centered around ∼1.5x inflation.
- **SEEDLingS** (LENA all-day audio, no on-camera observer): $\beta_{\text{react}} \sim \mathcal{N}(0, 0.001)$ — pinned at zero. Passive-recording channel, so no inflation.

- *Per-recording weighting.* Currently uniform; planned hook in `stan_data$log_r_obs_weight` for downweighting short or contaminated recordings. Off by default.

Stan model files form a partial order. Each file *adds* components on top of the previous, all toggleable via tight priors:

File	Adds	Used by
`log_irt.stan`	cross-sectional baseline ξ_i	Wordbank cross-sectional fits
`log_irt_long.stan`	admin→child indexing, ζ_i, bivariate prior	Wordbank longitudinal fits
`log_irt_io.stan`	per-recording $\log r^{\text{obs}}$ measurement layer, β_{react}	BabyView, SEEDLingS
`log_irt_proc.stan` *(planned)*	noisy-α observation from LWL RT/accuracy	Peekbank-linked Stanford datasets

The result is a small partial order of model files, not a combinatorial explosion. Adding a new dataset usually means writing a `prepare_<dataset>.R` script that reuses an existing Stan file with appropriate dataset-level constants; only when a genuinely new *observation channel* appears do we add a Stan file.

13 Comparison to prior accumulator models

Five prior models in this lineage all sit, with one or two parameterization changes, inside our nested family (§11). We work through each in chronological order, showing the mapping in our notation.

McMurray (2007) — pure deterministic accumulator

Each word j has a time-to-acquisition threshold τ_j drawn from a population distribution $g(\tau)$. At each unit time step every word accumulates one point, and word j is learned at the first t with $t \geq \tau_j$. The growth curve is the CDF of g:

$$L_{\text{McM}}(t) = \int_0^t g(\tau)\, d\tau = G(t), \quad L''(t) > 0 \iff g'(t) > 0.$$

Acceleration arises only if $g(\tau)$ puts more mass at moderately-hard ages than at easy ones. Frequency enters *implicitly* through whatever shapes $g(\tau)$; McMurray uses Gaussian by CLT.

Mapping. McMurray's deterministic accumulator is the zero-noise limit of our **M1** (unit-rate accumulator with no acceleration parameter). His τ_j corresponds to our log-threshold ψ_j on the time axis: in M1, the deterministic acquisition age for word j at population mean child is

$$t_j^{\text{M1}} = a_0 \cdot \exp(\psi_j - \log p_j - \log H - \mu_r),$$

which is the unique t where $\theta_{\text{pop}}(t) = \beta_j$. The distribution of $\{\psi_j\}$ across words plays the role of $g(\tau)$. Differences from McMurray:

1. *Stochastic observations* — we observe Bernoulli outcomes $y_{ij} \sim \text{Bernoulli}(\sigma(\theta - \beta))$; McMurray observes deterministic acquisition status. Our logit link is the smooth analogue of his step function.
2. *Per-child variation* — McMurray has none.
3. *Frequency in the linear predictor* — explicit $\log p_j$ enters the linear predictor; McMurray bundles it into $g(\tau)$.

Mitchell & McMurray (2009) — accumulator with feedback

Continuous-time generalization with a *leverage* feedback term c: the effective time argument $a(t)$ advances faster as the child's vocabulary $L(t)$ grows, so

$$a(t) = t + c\,L(t), \quad L(t) = G(t + c\,L(t)), \quad \frac{dL}{dt} = \frac{g(t + cL)}{1 - c\,g(t + cL)}.$$

Their key result: leverage shifts $L(t)$ earlier (or later, with $c < 0$) but *cannot create acceleration* that isn't already present in g. Acceleration still requires $g'(\tau) > 0$ over the relevant range. They also tested two frequency-to-difficulty mappings (additive and multiplicative); under Zipf with exponent 1, multiplicative gives uniform difficulties, hence linear growth.

Mapping. The feedback structure $a(t) = t + c\,L(t)$ does not have a clean analogue in our parameterization: it is intrinsically nonlinear in L. What we have instead is a *population-level* acceleration term $(1 + \delta)\,\log_{age}$ (M2) and *per-child* acceleration deviation ζ_i (M3), neither of which involves the child's own vocabulary state. Mitchell & McMurray's main *conclusion* — that acceleration must be in the model *somehow* — is operationalized in our family as the M2-vs-M1 comparison (RQ1). We confirm their conclusion empirically with $\hat{\delta} \gg 0$ and CrI excluding zero.

Hidaka (2013) — closed-form AoA distributions

Hidaka derives closed-form age-of-acquisition distributions for three nested learning processes:

Process	Dynamics	AoA distribution
Cumulative	N-accumulator, constant rate δ	$\Gamma(N, \delta)$
Rate-change	1-accumulator, rate $\delta\,t^D$	Weibull
Cumulative + rate-change	both	Weibull–Gamma

Closed-class words best fit Weibull (rate-change); content words best fit Gamma (cumulative). N correlates with frequency; change-of-rate exponent D correlates with imageability.

Mapping. Hidaka's three families correspond to three points in our nested family:

- *Cumulative* = our **M1**. Constant log-age slope; AoA for word j at population mean child has a one-point degenerate distribution. With per-child variance, AoA across kids is approximately log-normal.
- *Rate-change* = our **M2**. Slope on $\log_{age}$ is $1 + \delta$; on the linear time scale, effective tokens grow as $t^{1+\delta}$. Hidaka's exponent D corresponds (after reparameterization) to our δ.

- *Cumulative + rate-change* = our **M3**. Per-child intercept ξ_i and per-child slope deviation ζ_i jointly give a per-child power-law trajectory.

The functional families differ — Hidaka's AoA is Gamma/Weibull, ours is implicitly the convolution of a logistic CDF (item-level) with a normal (child-level) — but both live in the same "right-skewed positive AoA distribution" family. Our model gains explicit per-child structure (Hidaka fits population-averaged (N, δ)) and embedding in a Bayesian framework with real units.

Mollica & Piantadosi (2017) — Bayesian Gamma waiting time

Per-word generative model: accumulation begins at start age s_w; "effective learning instances" (ELIs) arrive as a Poisson process with rate λ_w; word is acquired after k_w ELIs. Acquisition age is therefore

$$\mathrm{AoA}_w \sim s_w + \Gamma(k_w, \lambda_w), \qquad F_w(a) = \int_0^{a-s_w} \frac{t^{k_w-1}\, e^{-t\lambda_w}\, \lambda_w^{k_w}}{\Gamma(k_w)}\, dt.$$

Aggregated data $x_{aw} \sim \mathrm{Bin}(F_w(a), N_a)$ is fit per-word with JAGS across 13 languages. They report $k \approx 10$, $\lambda \approx 0.5$/mo, $s \approx 2$ mo for comprehension.

Mapping. Mollica's per-word (k_w, λ_w) translates directly to our ψ_j and the population-level rate, but with two structural differences:

1. *Their model is per-word; ours is per-(word, child).* Mollica has no r_i or α_i. The total rate λ_w bundles input rate, child efficiency, and frequency-of-word-w into one number. We separate $\lambda \approx r_i \cdot \alpha_i \cdot p_j \cdot H$, with r_i externally pinned, enabling RQ3.
2. *Gamma waiting time vs. logistic acquisition curve.* With large k, the Gamma cumulative converges to a normal CDF (CLT); the logistic $\sigma(\theta-\beta)$ is its $\sim$1.7-scale-factor approximation. So with $k \gtrsim 5$ our Rasch model is approximately Mollica's Gamma waiting time, and ψ_j maps to $\log(k_w/\lambda_w)$ (the Gamma mean on the log scale).

Their joint (k, λ) ridge is the same identifiability problem as our (s, δ) ridge: both have one structural quantity (the mean / median acquisition time) and a near-orthogonal shape parameter that the data weakly constrains. Our resolution (externally pinning σ_r and per-class ψ priors) breaks the ridge in a different place than theirs.

Kachergis, Marchman, & Frank (2021) — IRT framing

Reframes the family as a 1-PL Rasch IRT model:

$$P(y_{ij} = 1 \mid \theta_i, d_j) = \frac{1}{1 + \exp(\theta_i + d_j)}.$$

Fit with `lme4::glmer` on Wordbank. Item-level covariates: expected tokens/month $(p_j \cdot H \cdot a)$, lexical class. Person-level covariate: age. Best-fit model `m3`: lexical class × tok_per_mo + age.

Mapping. Kachergis 2021 is essentially our M3 / M4 fit, but in standardized logit units with three structural simplifications:

1. *No external pin on σ_r.* Their θ_i absorbs both input rate and efficiency without any way to decompose them. RQ3 is unanswerable in their formulation.

2. *Age as additive person-covariate, not structural* $\log_{age}$. They write $\theta_i = \theta_{i,0} + \beta a$; we write $\theta_{it} = \xi_i + (1+\delta)\log((t-s)/a_0)$. The two are linearizations of each other for moderate age ranges, but ours preserves units (a slope of 1 on $\log_{age}$ is one log-token of evidence per log-age unit).

3. *No acceleration parameter.* Their model can't pose RQ1 quantitatively, although their finding that adding age to `m_tokmo` improves fit (i.e., age matters beyond expected tokens) is qualitatively the same conclusion.

Our nested family completes the standard-model program of Kachergis 2021 by (a) putting it in real token units rather than standardized logit units, (b) adding the structural δ for RQ1, (c) adding the external σ_r pin for RQ3, and (d) extending to input-uptake (BabyView, SEEDLingS) and processing (Peekbank-Stanford) datasets via the observation-channel architecture.

Summary table

Model	Per-child	Per-word	Frequency	Acceleration	Bayesian	Real units
McMurray 2007	–	$g(\tau)$	implicit	via g' only	–	–
Mitchell & McMurray 2009	–	$g(\tau) + c$	implicit	via g' only	–	–
Hidaka 2013	–	(N, δ, D)	via N	via D	–	–
Mollica & Piantadosi 2017	–	(k_w, λ_w)	implicit (in λ_w)	–	✓	partial
Kachergis et al. 2021	θ_i	d_j, class	explicit, unit-coef	–	–	–
Ours (M5)	$\xi_i, \zeta_i, \pi_\alpha$	ψ_j, λ_j, class	β_c free	δ free	✓	✓

A Start time s and the s/δ ridge

A natural fourth question — *when* does accumulation start? — is deferred to this appendix because the parameter s is near-non-identifiable in practice. Mollica & Piantadosi (2017) estimated $s \approx 2$ months for comprehension; we attempted a similar estimate for production but found the data does not support it, for the reason described below.

Identifiability in principle vs. practice. When s is freed jointly with δ, the two parameters trade off along a near-perfect ridge: posterior $r(s, \delta) \approx -0.96$ on both English and Norwegian longitudinal fits. In principle the pair is identifiable: setting $(1+\delta)\log(t-s) = (1+\delta')\log(t-s')$ for all t and differentiating in t gives $(1+\delta)/(t-s) = (1+\delta')/(t-s')$, forcing $\delta = \delta'$ and $s = s'$. Practically the curve-shape difference between two points along the ridge is small. For the English age range (16–30 mo), the ridge endpoints $(s, \delta) = (0.5, 9.4)$ and $(3, 8)$ produce $(1+\delta)\log((t-s)/a_0)$ curves that differ by only ~ 0.1 logits in shape (after subtracting a constant that ξ_i absorbs). With Bernoulli observations and finite N, that shape difference is at the noise floor. So s is *near-non-identifiable in practice*, and the reported s posterior in any `free_s` variant is only a coordinate choice along the ridge — it does not deliver an interpretable estimate of "onset of accumulation." We pin $s \approx 0$ in M1–M5 and treat `free_s` as a robustness check, not a first-class estimate.

Reference age a_0 is dataset-specific. a_0 is the pivot at which $\log_{age} = 0$ and so $\theta_{it} = \xi_i$ (per-child intercept). Setting a single global $a_0 = 20$ across datasets makes ξ_i language-relative in a misleading way: when admin ages skew far from a_0, the per-kid likelihood ridge in (ξ_i, ζ_i)

is tilted by an amount proportional to $\log(\bar{t}_i/a_0)$, pulling the marginal $r(\xi, \zeta)$ across kids in one direction or the other. We observed this as a spurious sign flip in $\rho_{\xi\zeta}$ between English (median admin age 19; $r = +0.39$) and Norwegian (median admin age 26; $r = -0.32$) on the same model. Re-evaluating each kid's ξ at their personal mid-observation age makes the flip vanish (Norwegian r becomes $+0.08$). We therefore set a_0 per dataset to the median admin age, in each `prepare_*.R` script. This is a reparameterization, not a model change: the likelihood, δ, and ζ_i are invariant; only the *interpretation* of ξ_i changes (from "ability at 20 mo" to "ability at the dataset's median admin age").